

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0896

Radnor 69 kV
September 26, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0896. The Interconnection Customer proposes a 158.8 MW solar photovoltaic facility interconnecting at the Radnor 69 kV substation in the North Carolina area. The request was received on November 23, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 283 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$13,550,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: CAPE HORN SUBSTATION – HUBEN 138 kV line
Contingency: HUTCHINS – BAUXITE 138 kV
Loading: 111.2% (Rating: 234 MVA)
DFAX: 5.9%
Pre-project loading: 107.2%
2. Facility: KERR DAM 345-138 kV XFMR
Contingency: TECUMSEH ENERGY CENTER – GASTON 345 kV
Loading: 104.8% (Rating: 507 MVA)
DFAX: 5.2%
Pre-project loading: 103.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HUBEN – CAPE HORN 138 kV Line	Rebuild line with higher-capacity conductor	\$20,234,000	47% (\$9,425,000)
KERR DAM 345-138 kV XFMR	Replace transformer with higher-rated unit	\$15,380,000	27% (\$4,125,000)
	Total allocated to project		\$13,550,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.